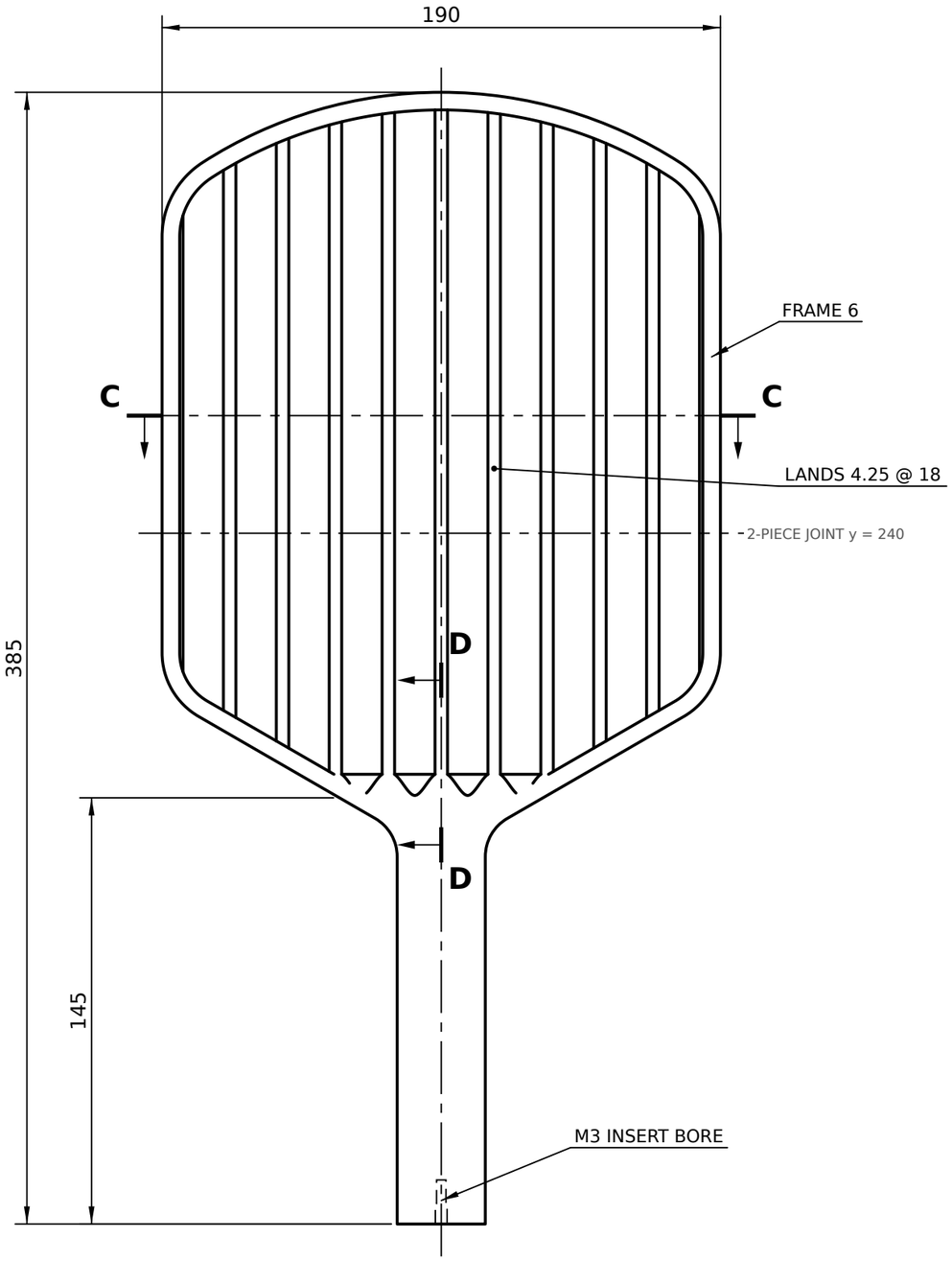
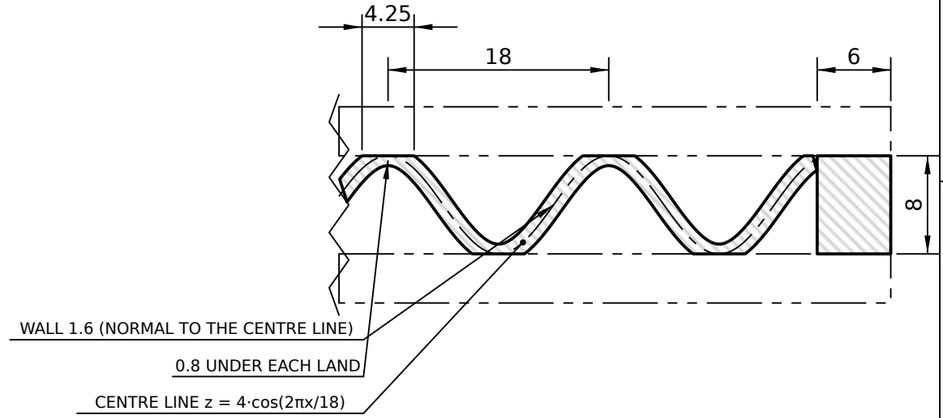


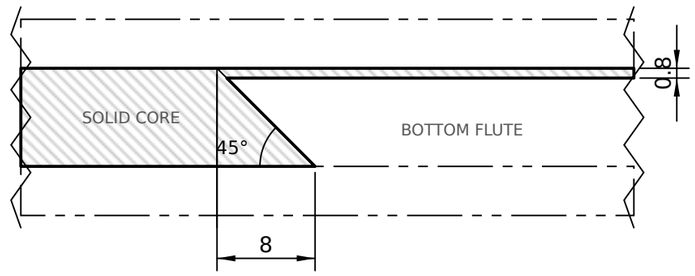
PLAN VIEW (TOP FACE)

SCALE 1:2 · BOTTOM LANDS OFFSET ½ PITCH



SECTION C–C (PART)

SCALE 2:1 · FACE SHEETS SHOWN PHANTOM



SECTION D–D (PART)

SCALE 2:1 · ON THE CENTRE LAND (x = 0), RUN-OUT FROM y = 150; TOP FLUTES MIRROR IT

NOTES

1. MATERIAL: PA-CF (CARBON-FIBRE-FILLED NYLON), E ≈ 4 GPa, ρ 1.18 g/cm³. DRY THE FILAMENT BEFORE PRINTING.
2. PRINT FLAT, 100 % SOLID: WALL 1.6 = 4 PERIMETERS (0.4 NOZZLE), 0.2 LAYERS. NO SUPPORTS (45° RUN-OUTS).
3. THICKNESS 8 ±0.1. BOTH FACES ARE BOND SURFACES, FLAT WITHIN 0.1.
4. WAVE RIB λ18, WALL 1.6, CENTRE-LINE AMPLITUDE 4; CRESTS CUT FLAT = LANDS 4.25 ON BOTH FACES.
5. FRAME 6, CLOSED. SOLID TO y = 150; EACH FLUTE STARTS WITH A 45° RUN-OUT, 8 LONG (D–D).
6. M3 HEAT-SET INSERT BORE Ø4 × 7 + Ø3.2 × 15 IN THE BUTT FACE (GRIP SCREW) – SEE PB-GR-002.
7. BONDING: ABRADE (P120), DEGREASE, STRUCTURAL EPOXY, FILLET EVERY LAND EDGE.
8. PRINT FILES output/rib_support/: hybrid_rib_support_1piece.stl (BED ≥ 391 × 196) OR _2piece_A + _B (256 BED): 7 SOLID SPLICE BAND AT y = 240, DOVETAIL JOINT 0.1 CLEARANCE PER SIDE - GLUE IT BEFORE BONDING THE SHEETS.
9. MASS 170 g. y FROM THE GRIP BUTT FACE (y = 0); THE RIB SUPPORT STARTS AT y = 5.

RIB SUPPORT – HYBRID

RIB SUPPORT (CORE) · 3D-PRINTED PA-CF · 385 × 190 × 8 mm

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-24		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	C	3 / 3	A3	1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PA-CF (NOTE 1)	ISSUED FOR REVIEW